Infant formula (ready-to-feed, liquid) — HMTc certification standard

Institute of Contaminant Standards — Heavy Metal Tested & Certified program

Version 1.0 · Native basis: as_consumed · 8/8 analytes with a published limit

STANDARD

infant-formula-rtf-liquid, version 1.0. HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

PROVENANCE

Compiled from data/evidence/regulatory_limits.csv. Each limit is a lookup against sovereign law, not an occurrence computation.

TAMPER-EVIDENT SEAL

The SHA-256 of this exact PDF is sealed by an RFC-3161 trusted timestamp. The token (standard.pdf.tsr) and the manifest (manifest.json) accompany this file; the date of record is the timestamping authority's, not the program's.

Abstract

This is the frozen, version-pinned Heavy Metal Tested & Certified standard for infant formula (ready-to-feed, liquid). Each per-analyte certification limit is the strictest heavy-metal maximum level set by any government in the world for this product class, expressed in the product's native basis (as_consumed) and derived by lookup against sovereign law under HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The per-jurisdiction derivation below is the complete "show your work". This snapshot is immutable: any material change to a limit mints a new version with its own timestamped snapshot.

1 | Certification limits

All values in ppb (µg/kg), as_consumed basis. A brand meeting every published value on a like-for-like basis qualifies for the mark. Tier 1 toxics (Pb, Cd, iAs, mercury) carry a zero-tolerance posture; Tier 2 metals (Ni, Al, Sn, total Cr) carry the program's transitional allowance. Where a limit is set in another basis by the binding jurisdiction it is converted into the native basis and the conversion is named in the "Basis derivation" column.

Cow-milk based

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	10	European Union	same basis	regulatory-alignment
Inorganic arsenic (iAs)	Tier 1	10	European Union	same basis	regulatory-alignment
Cadmium (Cd)	Tier 1	5	European Union	same basis	regulatory-alignment

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Mercury (tHg)	Tier 1	4	Israel	same basis	regulatory-alignment
Nickel (Ni)	Tier 2	100	European Union	same basis	regulatory-alignment
Aluminium (Al)	Tier 2	500	Australia / NZ	same basis	regulatory-alignment
Tin (Sn)	Tier 2	5000	India	face value	regulatory-alignment
Total chromium (Cr)	Tier 2	300	—	face value (nearest government value; comparator vs any direct limit)	read_across

Soy based

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	10	European Union	same basis	regulatory-alignment
Inorganic arsenic (iAs)	Tier 1	10	European Union	same basis	regulatory-alignment
Cadmium (Cd)	Tier 1	5	Israel	same basis	regulatory-alignment
Mercury (tHg)	Tier 1	4	Israel	same basis	regulatory-alignment
Nickel (Ni)	Tier 2	100	European Union	same basis	regulatory-alignment
Aluminium (Al)	Tier 2	1000	Australia / NZ	same basis	regulatory-alignment
Tin (Sn)	Tier 2	5000	India	face value	regulatory-alignment
Total chromium (Cr)	Tier 2	300	—	face value (nearest government value; comparator vs any direct limit)	read_across

2 | Per-jurisdiction derivation

The binding value for each analyte is the lowest sovereign maximum level across the jurisdictions on record, in the native basis. Rows marked "converted" were set by the jurisdiction in the other basis and carried across the reconstitution ratio; rows marked "native" were already in this basis. Every value is keyed to its record in the sovereign register.

Cow-milk based

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	Australia / NZ	10	native	aunz_sch19_pb_formula_10
—	Brazil	10	native	br_pb_pediatric-formula-for-enteral--nutrition_10
—	Canada	10	native	ca_pb_infant-formula-all-types-two-former--limi_10

—	China	10	native	cn_pb_liquid-ready-to-feed-infant-formula_-10
—	Codex Alimentarius	10	native	codex_pb_infant-formula-formula-for--special-medic_10
—	European Union-(binding)	10	native	eu2023_pb_formula_liquid_10
—	Gulf (GSO)	10	native	gso193_2021_pb_infant_formula_formula_for_-special_medical_purposes_intended_for_-infants_10
—	HK	10	native	hk_cap132v_pb_infant-formula-and-follow--up-formula_10
—	Israel	10	native	il_fcs2016_pb_15_infant_formula_ready_to_-eat_and_follow_up_formula_10
—	South Korea	10	native	kr_pb_infant-formula-follow-up-formula--cereal_10
—	Singapore	10	native	sg_foodregs_2026_infant_formula_as_-consumed_pb_10
—	Thailand	10	native	th_moph414_2020_pb_formula_as_consumed_10
—	United Kingdom	10	native	uk_pb_infant-formulae-and-follow-on--formulae-m_10
—	Colombia	20	native	co_res4506_2013_pb_formula_ready_20
—	Fiji	20	converted	fj_fsr2009_pb_006_infant_formula_20
—	India	20	native	in_pb_infant-formula-ready-to-use_20
—	VN	20	native	vn_qcvn8_2_2011_pb_formulae-for-young--children-aged-under-36-mont_20
iAs	European Union-(binding)	10	native	eu2023_ias_formula_liquid_10
—	Israel	10	native	il_fcs2016_ias_14_infant_and_toddler_-formulas_ready_to_eat_10
—	Brazil	20	native	br_ias_pediatric-formula-for-enteral--nutrition_20
—	South Korea	100	converted	kr_ias_infant-formula-follow-up-formula--cereal_100
Cd	European Union-(binding)	5	native	eu2023_cd_formula_cow_liquid_5
—	Gulf (GSO)	5	native	gso193_2021_cd_liquid_formulae_-manufactured_from_cows_milk_proteins_or_-protein_hydrolys_5
—	Israel	5	native	il_fcs2016_cd_14_infant_and_toddler_-formulas_ready_to_eat_5
—	United Kingdom	5	native	uk_cd_infant-formulae-and-follow-on--formulae-l_5

—	Brazil	10	native	br_cd_pediatric-formula-for-enteral--nutrition_10
—	India	100	converted	in_cd_infant-milk-substitute-and-infant--foods_100
tHg	Israel (binding)	4	native	il_fcs2016_thg_14_infant_and_toddler_-formulas_ready_to_eat_4
Ni	European Union- (binding)	100	native	eu2024_ni_formula_liquid_100
Al	Australia / NZ (binding)	500	native	aunz_sch19_al_formula_nonsoy_500
Sn	India (binding)	5000	converted	in_sn_infant-milk-substitute-and-infant--foods_5000

Soy based

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	Australia / NZ	10	native	aunz_sch19_pb_formula_10
—	Brazil	10	native	br_pb_pediatric-formula-for-enteral--nutrition_10
—	Canada	10	native	ca_pb_infant-formula-all-types-two-former--limi_10
—	China	10	native	cn_pb_liquid-ready-to-feed-infant-formula_-10
—	Codex Alimentarius	10	native	codex_pb_infant-formula-formula-for--special-medic_10
—	European Union- (binding)	10	native	eu2023_pb_formula_liquid_10
—	Gulf (GSO)	10	native	gso193_2021_pb_infant_formula_formula_for_-special_medical_purposes_intended_for_-infants_10
—	HK	10	native	hk_cap132v_pb_infant-formula-and-follow--up-formula_10
—	Israel	10	native	il_fcs2016_pb_15_infant_formula_ready_to_-eat_and_follow_up_formula_10
—	South Korea	10	native	kr_pb_infant-formula-follow-up-formula--cereal_10
—	Singapore	10	native	sg_foodregs_2026_infant_formula_as_-consumed_pb_10
—	Thailand	10	native	th_moph414_2020_pb_formula_as_consumed_10
—	United Kingdom	10	native	uk_pb_infant-formulae-and-follow-on--formulae-m_10
—	Colombia	20	native	co_res4506_2013_pb_formula_ready_20
—	Fiji	20	converted	fj_fsr2009_pb_006_infant_formula_20

—	India	20	native	in_pb_infant-formula-ready-to-use_20
—	VN	20	native	vn_qcvn8_2_2011_pb_formulae-for-young--children-aged-under-36-mont_20
iAs	European Union-(binding)	10	native	eu2023_ias_formula_liquid_10
—	Israel	10	native	il_fcs2016_ias_14_infant_and_toddler_-formulas_ready_to_eat_10
—	Brazil	20	native	br_ias_pediatric-formula-for-enteral--nutrition_20
—	South Korea	100	converted	kr_ias_infant-formula-follow-up-formula--cereal_100
Cd	Israel (binding)	5	native	il_fcs2016_cd_14_infant_and_toddler_-formulas_ready_to_eat_5
—	Brazil	10	native	br_cd_pediatric-formula-for-enteral--nutrition_10
—	European Union	10	native	eu2023_cd_formula_soy_liquid_10
—	Gulf (GSO)	10	native	gso193_2021_cd_liquid_formulae_-manufactured_from_soya_protein_isolates_-alone_or_in_a_mi_10
—	United Kingdom	10	native	uk_cd_infant-formulae-and-follow-on--formulae-l_10
—	India	100	converted	in_cd_infant-milk-substitute-and-infant--foods_100
tHg	Israel (binding)	4	native	il_fcs2016_thg_14_infant_and_toddler_-formulas_ready_to_eat_4
Ni	European Union-(binding)	100	native	eu2024_ni_formula_liquid_100
Al	Australia / NZ (binding)	1000	native	aunz_sch19_al_formula_soy_1000
Sn	India (binding)	5000	converted	in_sn_infant-milk-substitute-and-infant--foods_5000

3 | Method and provenance

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The default standard for any (product, analyte) is the strictest maximum level set by a credible government regulator, converted to the row's native basis. Where the law itself splits a limit by formulation, the standard adopts the law's own split and cites it. Species limits are bounded by their parent total (iAs ≤ tAs, MeHg ≤ tHg, Cr-VI ≤ total Cr); read-across limits (for example aluminium carried from an as-consumed limit to an as-sold basis across the reconstitution ratio) are disclosed as read-across and never presented as direct governance.

The occurrence and toxicology corpus that contextualises these limits is maintained independently as the Heavy Metal Index literature baseline (heavymetalindex.com). This certification standard cites that baseline one way; it does not use the baseline's contents as the basis for any government-limit value stated here.

4 | Version and tamper-evidence

This is version 1.0 of the Infant formula (ready-to-feed, liquid) standard. A certification number becomes final when its version carries a resolvable DataCite DOI; a material change to any limit mints a new version with its own frozen snapshot, so a superseded number is never silently overwritten. Tamper-evidence does not depend on trusting the program: the SHA-256 of this exact PDF was submitted to an RFC-3161 trusted timestamping authority (https://freetsa.org/tsr), which returned a signed token binding that hash to a time it — not the program — attests. The token (standard.pdf.tsr), the authority's CA chain (tsa-cacert.pem), and the manifest (manifest.json) travel with this file.

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